LECTURE 16

SOFTWARE QUALITY MANAGEMENT AND CONFIGURATION MANAGEMENT

Knowledge Area	Process				
	Initiating	**Planning**	**Executing**	**Monitoring & Control**	**Closing**
Quality Management		Quality Planning	Perform Quality Assurance	Perform Quality Control	

What is Quality Management?

- Quality Management on a project involves the **processes** and **activities** that ensure the project and product requirements of the project are **validated**, **checked** and **fulfilled**.

- It refers to the creation of **policies** and **procedures** that are followed during the project to ensure that the **project meets the needs required by the customer**.

Quality Management Processes

- Plan Quality Management
- Perform Quality Assurance
- Control Quality

What is quality?

Quality is the measurement of how closely your product meets its requirements.

Quality vs. Grade:

Quality means that something does what you needed it to do. Grade describes how much people value it.

What is Software Quality

Software quality is defined as a field of study and practice that describes the desirable attributes of software products.

There are two main approaches to software quality:

1. **Defect management**
2. **Quality attributes**

Software Quality Defect Management Approach

- A software defect can be regarded as any failure to address end-user requirements. Common defects include missed or misunderstood requirements and errors in design, functional logic, data relationships, process timing, validity checking, and coding errors.

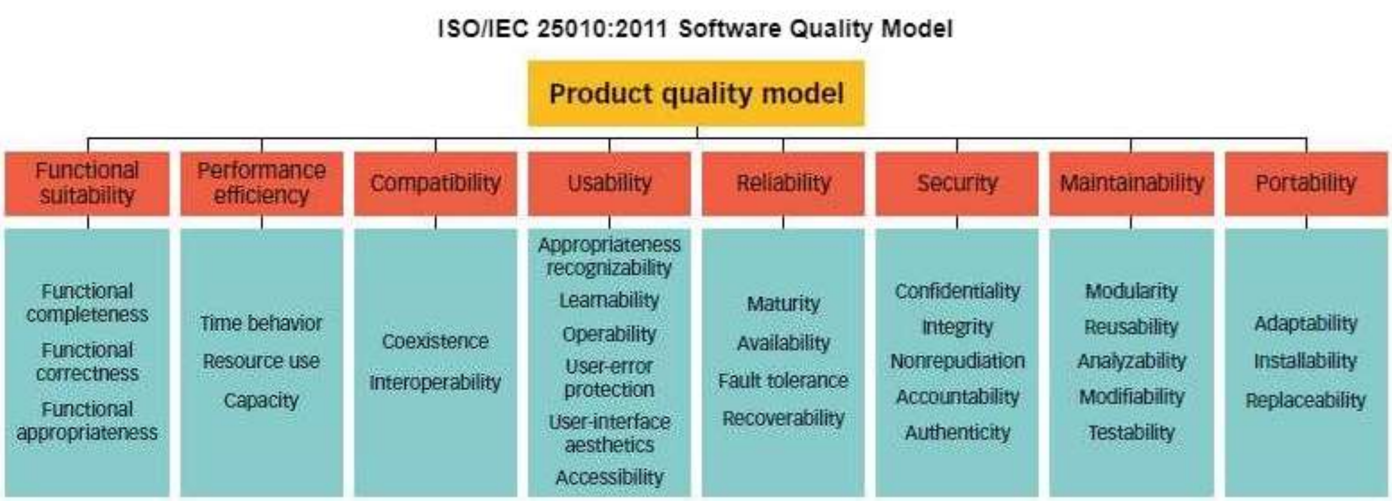
- The software defect management approach is based on counting and managing defects. Defects are commonly categorized by severity, and the numbers in each category are used for planning.

- More mature software development organizations use tools, such as defect leakage matrices (for counting the numbers of defects that pass through development phases prior to detection) and control charts, to measure and improve development process capability.

Software Quality Attributes Approach

This approach to software quality is best exemplified by fixed quality models, such as ISO/IEC 25010:2011. This standard describes a hierarchy of eight quality characteristics, each composed of sub-characteristics:

- Functional suitability
- Reliability
- Operability
- Performance efficiency
- Security
- Compatibility
- Maintainability
- Transferability



Relationship between Quality Assurance, Quality Control and Testing

	QA	QC	Testing
Purpose	Setting up adequate processes, introducing the standards of quality to prevent the errors and flaws in the product	Making sure that the product corresponds to the requirements and specs before it is released	Detecting and solving software errors and flaws
Focus	Processes	Product as a whole	Source code and design
What	Prevention	Verification	Detection
Who	The team including the stakeholders	The team	Test Engineers, Developers
When	Throughout the process	Before the release	At the testing stage or along with the development process

How can you ensure quality on the project?

- **Define** what an **acceptable level of quality** on the project is.
- **Determine** what **metrics** will be used to **measure quality** before actual work on the project begins.
- **Inspect work as it is done**, not afterwards.
- Check the quality of **activities** and **work packages** before they are completed.
- Determine what is the desired level of performance in the product?
- **Measure** actual **performance** against defined metrics

SOME ESSENTIAL QUALITY CONCEPT:

Customer satisfaction:

- It is about making sure that the people who are paying for the end product are happy with what they get.

- When the team gathers requirements for the specification, they try to write down all of the things that the customers want in the product so that you know how to make them happy.

- Some requirements can be left **unstated**, too. Those are the ones that are implied by the customer's explicit needs. In the end, if you fulfill all of your requirements, your customers should be really satisfied.

Fitness for use:

- It is about making sure that the product you build has the best design possible to fit the customer's needs. Which would you choose: a product that's beautifully designed, well-constructed, solidly built, and all-around pleasant to look.

- Quality is "fitness for use." A high-quality product does what its customers want in such a way that they actually use the product.

Conformance to requirements:

- It is the core of both customer satisfaction and fitness for use. Above all, your product needs to do what you wrote down in your requirements specification. Your requirements should take into account both what will satisfy your customer and the best design possible for the job.

- Your product's quality is judged by whether you built what you said you would build.

Avoiding gold-plated software:

- You create *gold-plated software* when you complete a project, and the software is ready to go to the customer, but suddenly realize that you have money to burn.

- If you find yourself with a hefty sum of cash remaining in the project budget, you may feel tempted to fix the situation with a lot of bling.

Who is responsible for Quality?

The **project manager** is responsible for Quality. However, the team members should check their work before submitting it forward to ensure that it meets the requirements.

W.Edwards Deming:

Quality Expert Edwards Deming said that 85% of quality problems are because of the **management environment** and the **system** in which the team **performs** their work.

Effects of Poor Quality on a Project	Effects of Good Quality on a Project
• Rework • Higher costs • Higher risk • Delay in schedule • Low customer satisfaction • Lower morale of team	• No rework • Lesser costs • Lesser risk • Completing work ahead of schedule • High customer satisfaction • Higher morale of team

Plan Quality management Process:

Plan Quality Management primarily comprises of the following:

1. **Identifying requirements and existing standards**. This includes industry practices, processes, standards and requirements for determining the quality of the project.

2. **Planning** how you can **meet the quality standards** and **requirements**.

Plan Quality Management Process Overview:

1. **Organization Process Assets:**
   - The project must comply with existing standards, so the project manager first identifies all existing standards, policies and procedures, which may be in the form of Organization Process Assets (Internal).
2. **Enterprise Environmental Factors:**
   - Then, the Project Manager needs to identify all external practices and standards, which may be in the form of Enterprise environmental factors (external).
3. **Customer's Quality standards and requirements:**
   - Finally, the project manager identifies all the customer requirements and quality standards.
4. **Create Project Specific Standards**
   - Once existing standards and requirements have been identified, the project manager then creates project specific standards and procedures
5. **Define Quality and how it will be achieved:**
   - The PM then defines how quality will be achieved on the project, keeping in view the customer's requirements and the existing standards.
   - Defining these measures of quality will allow the project manager to evaluate the work as it is being done and know whether it is out of control and when changes are required.

Plan Quality Management - Tools and Techniques:

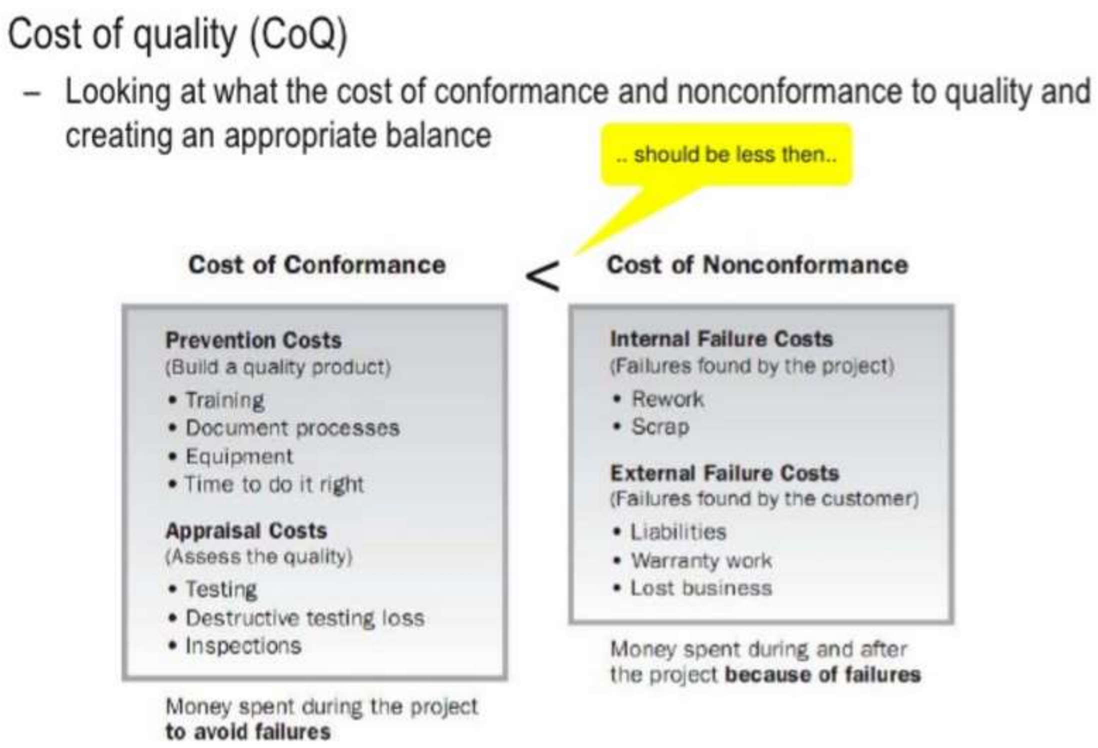
Cost-benefit analysis is looking at how much your quality activities will cost versus how much you will gain from doing them. The costs are easy to measure; the effort and resources it takes to do them are just like any other task on your schedule. Since quality activities don't actually produce a product, though, it is harder for people to measure the benefits sometimes. The main benefits are less rework, higher productivity and efficiency, and more satisfaction from both the team and the customer.

Benchmarking means using the results of Plan Quality on other projects to set goals for your own. You might find that the last project your company did had 20% fewer defects than the one before it. You would want to learn from a project like that, and put in practice any of the ideas the company used to make such a great improvement.
Benchmarks can give you some reference points for judging your own project before you even get started with the work.

Design of experiments is where you apply the scientific method to create a set of tests for your project's deliverables. It's a *statistical* method, which means you use statistics to analyze the results of your experiments to determine how your deliverables best meet the requirements. A lot of quality managers use this technique to produce a list of tests that they'll run on the deliverables, so they have data to analyze later.

Meetings are used to figure out how your team will do all of the quality-related activities your project requires. The whole team might collaborate on the Quality Management plan in these meetings.

Statistical sampling is when you look at a representative sample of something to make decisions. For example, you might take a look at a selection of widgets produced in a factory to figure out which quality activities would help you prevent defects in them.



Perform Quality Assurance Process:

The process of **auditing the quality requirement and the result of quality control measurements** to ensure appropriate quality standards and operational definitions are used.

Are we using the standard?
Can we improve the standard?

Quality Audits

- To see if you are complying with company policies, standards & procedures
- Determine whether they are used efficiently & effectively
- Identify all the good practices being implemented
- Identify all the gaps/shortcomings
- Look for new lesson learned & good practices

Process Analysis

- Includes root cause analysis

Control Quality Process:

- Control Quality is about ensuring that the deliverables meet the quality standards.
- It is the Monitoring and Controlling process where you look at each deliverable and inspect it for defects.

- Metrics make it easy for you to check how well your product meets expectations.

Control Quality - Tools and Techniques:

- Quality Tools
- Software Testing

Quality Tools:

- **Cause and Effect Diagram (Ishikawa Diagram or Fishbone Diagram)**
  - Helps stimulate thinking, organize thoughts, and generate discussion
  - Can be used to explore the factors that will result in a desire future outcome

- **Histogram**
  - Showing how often a particular problem/situation occurred

- **Pareto Chart/Diagram (80/20 principle)**
  - Histogram ordered by frequency of occurrence which used to focus attention on the most critical issues
  - 80% of the problems are due to 20% of the causes
- **Run Chart**
  - To look at history and see a pattern of variation

- **Scatter Diagram**
  - Regression analysis

Software Testing:

- **Software testing** is a process of executing a program or application with the intent of finding the **software** bugs.

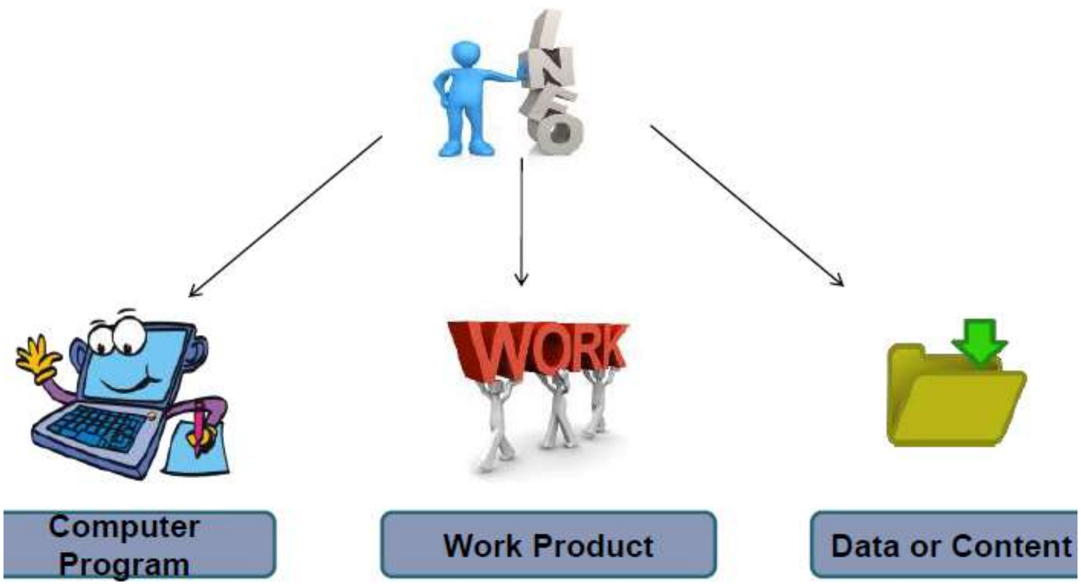
- It can also be stated as the process of validating and verifying that a **software** program or application or product: Meets the business and technical requirements that guided its design and development.

- There are many different types of **software testing** but the two main categories are dynamic **testing** and static **testing**.

SOFTWARE CONFIGURATION MANAGEMENT

What is Software Configuration Management?

- Software Configuration Management (SCM) is a process to systematically manage, organize, and control the changes in the documents, codes, and other entities during the Software Development Life Cycle.
- The primary goal is to increase productivity with minimal mistakes.
- **Output of software process is information. Information is divided into these parts.**



Definition:
The items that comprise all information produced as a part of software process is called **software configuration.**

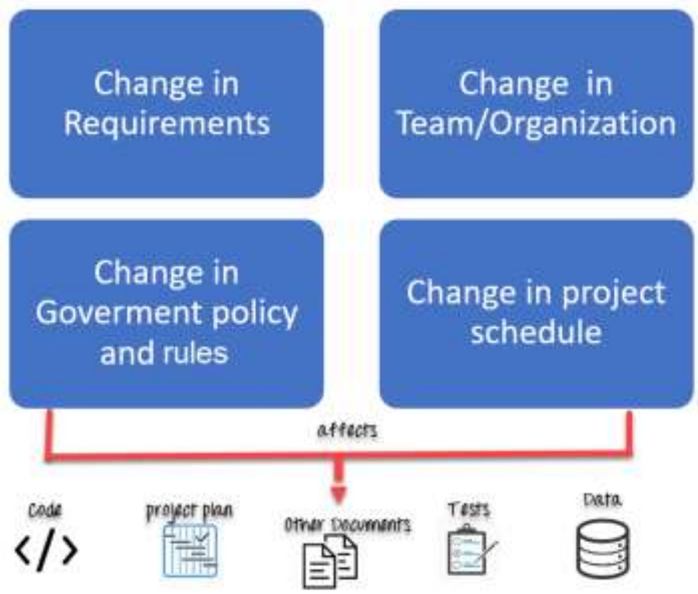
First Law of System Engineering
"No matter where you are in the system life cycle, the system will change, and the desire to change it will persist throughout the life cycle."

Changes occur because of
- New business or market conditions.
- New stakeholder needs demand modification of data produced by information systems.
- Reorganization or business growth/downsizing.
- Budgetary or scheduling constraints cause a redefinition of the system or product.

Why do we need Configuration management?

The primary reasons for Software Configuration Management System are:

- There are multiple people working on software which is continually updating
- It may be a case where multiple version, branches, authors are involved in a software config project, and the team is geographically distributed and works concurrently
- Changes in user requirement, policy, budget, schedule need to be accommodated.
- Software should able to run on various machines and Operating Systems
- Helps to develop coordination among stakeholders
- SCM process is also beneficial to control the costs involved in making changes to a system



Any change in the software configuration Items will affect the final product. Therefore, changes to configuration items need to be controlled and managed.

Tasks in SCM process

- Configuration Identification
- Baselines
- Change Control
- Configuration Status Accounting
- Configuration Audits and Reviews

Configuration Identification:

Configuration identification is a method of determining the scope of the software system. With the help of this step, you can manage or control something even if you don't know what it is. It is a description that contains the CSCI type (Computer Software Configuration Item), a project identifier and version information.

Activities during this process:

- Identification of configuration Items like source code modules, test case, and requirements specification.
- Identification of each CSCI in the SCM repository, by using an object-oriented approach
- The process starts with basic objects which are grouped into aggregate objects. Details of what, why, when and by whom changes in the test are made
- Every object has its own features that identify its name that is explicit to all other objects
- List of resources required such as the document, the file, tools, etc.

Example:
Instead of naming a File login.php its should be named login_v1.2.php where v1.2 stands for the version number of the file
Instead of naming folder "Code" it should be named "Code_D" where D represents code should be backed up daily.

Baseline:
A baseline is a formally accepted version of a software configuration item. It is designated and fixed at a specific time while conducting the SCM process. It can only be changed through formal change control procedures.

Activities during this process:

- Facilitate construction of various versions of an application
- Defining and determining mechanisms for managing various versions of these work products
- The functional baseline corresponds to the reviewed system requirements
- Widely used baselines include functional, developmental, and product baselines

In simple words, baseline means ready for release.

Change Control:
Change control is a procedural method which ensures quality and consistency when changes are made in the configuration object. In this step, the change request is submitted to software configuration manager.
Activities during this process:

- Control ad-hoc change to build stable software development environment. Changes are committed to the repository
- The request will be checked based on the technical merit, possible side effects and overall impact on other configuration objects.
- It manages changes and making configuration items available during the software lifecycle

Configuration Status Accounting:
Configuration status accounting tracks each release during the SCM process. This stage involves tracking what each version has and the changes that lead to this version.

Activities during this process:

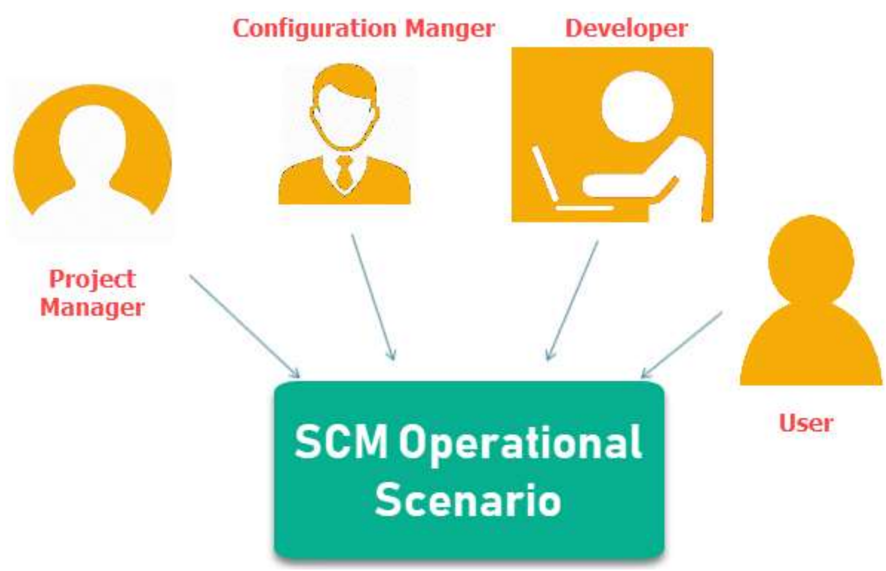
- Keeps a record of all the changes made to the previous baseline to reach a new baseline
- Identify all items to define the software configuration
- Monitor status of change requests
- Complete listing of all changes since the last baseline
- Allows tracking of progress to next baseline
- Allows to check previous releases/versions to be extracted for testing

Configuration Audits and Reviews:
Software Configuration audits verify that all the software product satisfies the baseline needs. It ensures that what is built is what is delivered.

Activities during this process:

- Configuration auditing is conducted by auditors by checking that defined processes are being followed and ensuring that the SCM goals are satisfied.
- To verify compliance with configuration control standards. auditing and reporting the changes made
- SCM audits also ensure that traceability is maintained during the process.
- Ensures that changes made to a baseline comply with the configuration status reports
- Validation of completeness and consistency

Participant of SCM process:
Following are the key participants in SCM



1. Configuration Manager

- Configuration Manager is the head who is Responsible for identifying configuration items.
- CM ensures team follows the SCM process
- He/She needs to approve or reject change requests

2. Developer

- The developer needs to change the code as per standard development activities or change requests. He is responsible for maintaining configuration of code.
- The developer should check the changes and resolves conflicts

3. Auditor

- The auditor is responsible for SCM audits and reviews.
- Need to ensure the consistency and completeness of release.

4. Project Manager:

- Ensure that the product is developed within a certain time frame
- Monitors the progress of development and recognizes issues in the SCM process
- Generate reports about the status of the software system
- Make sure that processes and policies are followed for creating, changing, and testing

5. User

The end user should understand the key SCM terms to ensure he has the latest version of the software

Software Configuration Management Plan

The SCMP (Software Configuration management planning) process planning begins at the early coding phases of a project. The outcome of the planning phase is the SCM plan which might be stretched or revised during the project.

- The SCMP can follow a public standard like the IEEE 828 or organization specific standard
- It defines the types of documents to be management and a document naming. Example Test_v1
- SCMP defines the person who will be responsible for the entire SCM process and creation of baselines.
- Fix policies for version management & change control
- Define tools which can be used during the SCM process
- Configuration management database for recording configuration information.

Software Configuration Management Tools

Any Change management software should have the following 3 Key features:

Concurrency Management:

When two or more tasks are happening at the same time, it is known as concurrent operation. Concurrency in context to SCM means that the same file being edited by multiple persons at the same time.

If concurrency is not managed correctly with SCM tools, then it may create many pressing issues.

Version Control:
SCM uses archiving method or saves every change made to file. With the help of archiving or save feature, it is possible to roll back to the previous version in case of issues.

Synchronization:
Users can checkout more than one files or an entire copy of the repository. The user then works on the needed file and checks in the changes back to the repository.They can synchronize their local copy to stay updated with the changes made by other team members.
Following are popular tools

1. Git: Git is a free and open source tool which helps version control. It is designed to handle all types of projects with speed and efficiency.
Download link: https://git-scm.com/

2. Team Foundation Server: Team Foundation is a group of tools and technologies that enable the team to collaborate and coordinate for building a product.
Download link: https://azure.microsoft.com/en-us/services/devops/server/